dehydrogenase. The study of enzyme is called Enzymology.

Enzymes are proteins or large biomolecules that can catalyze certain biochemical reactions for metabolic process within the cell. The substances that can speed up a chemical reaction without being permanently altered itself are called catalysts. Enzymes accelerate the rate of chemical reactions. The molecule upon which enzyme may act are called substrate and the enzyme convert the substrate into different molecules known as products. The enzyme serves as biological catalyst (Table 4.3).



Tyrosinases are synthesized by *Agaricus bisporus,* which is involved in melanogenesis (pigmentation of skin and hair).

4.10.1 Characteristics of Enzymes

Enzymes

- are highly substrate specific
- are reused at several times
- synthesized within the cells are determined by genes
- speed up the chemical reaction
- decrease the activation energy needed to start
- act as a biocatalyst

Infobits

Proteins have four levels of structure (i) primary (sequence of amino acids), (ii) secondary (regular coils or pleats linked by peptide bonds), (iii) tertiary overall three dimensional structure of a polypeptide linked by disulphide bonds) and (iv) quarterly structure (two or more polypeptides chains). Like all proteins, enzymes are composed of one or more long chain of inter connected amino acids.

Table 4.3: Enzyme Classification Based on Type of Chemical Reaction

Class	Type of Chemical Reaction	Reactions	Examples
Oxido-reductase	Oxidation-reduction in which oxygen and hydrogen are gained or lost	$Xr_{ed} + Y_{ox} \longrightarrow X_{ox} + Y_{re}d$	Cytochrome oxidase, lactate dehydrogenase
Transferase	Transfer of functional groups, such as an amino group, acetyl group, or phosphate group	$X - P + Y \longrightarrow X + Y - P$	Acetate kinase, alanine deaminase, transaminase, phosphotransferase
Hydrolase	Hydrolysis (addition of water)	$X - Y + H_2O \longrightarrow X - H + Y - OH$	Lipase, sucrose
Lyase	Removal of groups of atoms without hydrolysis	$X - Y \longrightarrow X + Y$	Oxalate decarboxylase, isocitrate, lyase
Isomerase	Rearrangement of atoms within a molecule	$X - Y - Z \longrightarrow X - Z - Y$	Glucose-phosphate isomerase, alanine racemase
Ligase	Joining of two molecules (using energy usually derived from breakdown of ATP)	$X + Y + ATP \longrightarrow X - Y + ADP + pi$	Acetyl-CoA synthetase, DNA ligase

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